

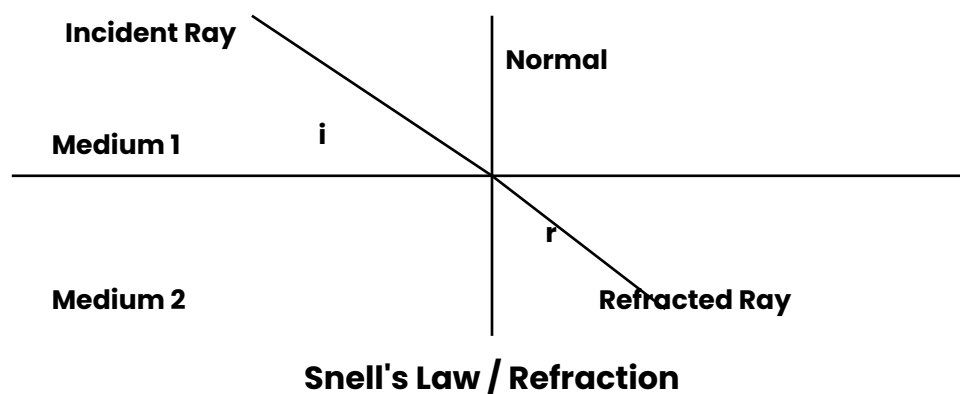
REFRACTION

Refraction of Light

1. Refraction of Light

Refraction is the change in direction of light when it passes obliquely from one transparent medium to another.

- The speed of light changes when it enters a different medium.
- The frequency of light remains unchanged during refraction.
- The wavelength changes when the speed of light changes.



2. Laws of Refraction — Snell's Law

- The incident ray, refracted ray and normal at the point of incidence lie in the same plane.
- For a given pair of media, the ratio of sine of angle of incidence to sine of angle of refraction is constant.

$$\sin i / \sin r = \text{constant}$$

$$n = \sin i / \sin r$$

3. Refractive Index

Refractive index measures how much a medium slows down and bends light compared with vacuum.

$$n = c / v$$

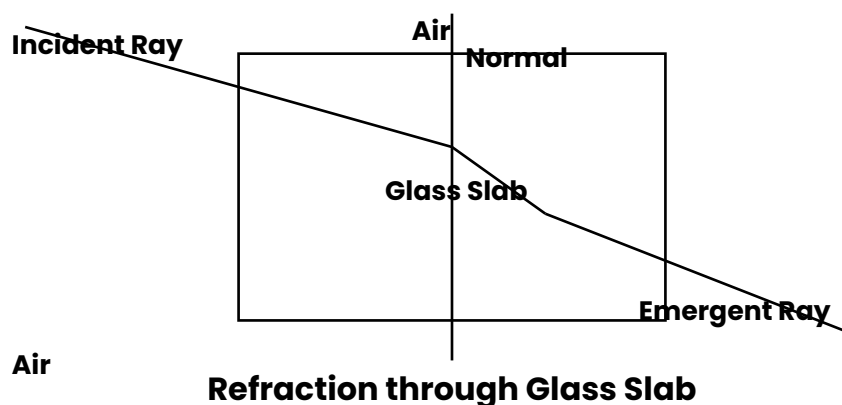
- c is the speed of light in vacuum.
- v is the speed of light in the medium.

$$n_{21} = v_1 / v_2$$

4. Refraction through Glass Slab

When a ray of light passes through a plane parallel glass slab, it undergoes refraction at both surfaces.

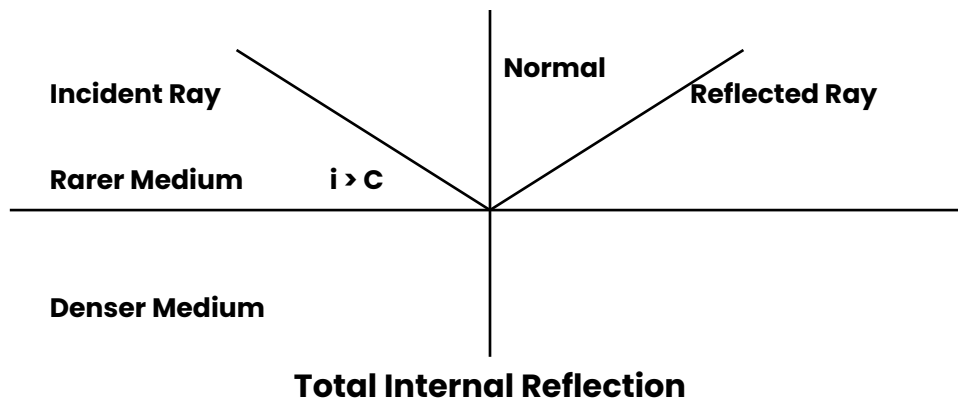
- The emergent ray is parallel to the incident ray.
- The emergent ray is laterally displaced from the incident ray.
- The amount of lateral displacement depends on the thickness of the slab, angle of incidence and refractive index.



5. Total Internal Reflection (TIR)

Total internal reflection is the complete reflection of light back into a denser medium when it travels towards a rarer medium under suitable conditions.

- Light must travel from a denser medium to a rarer medium.
- The angle of incidence must be greater than the critical angle.

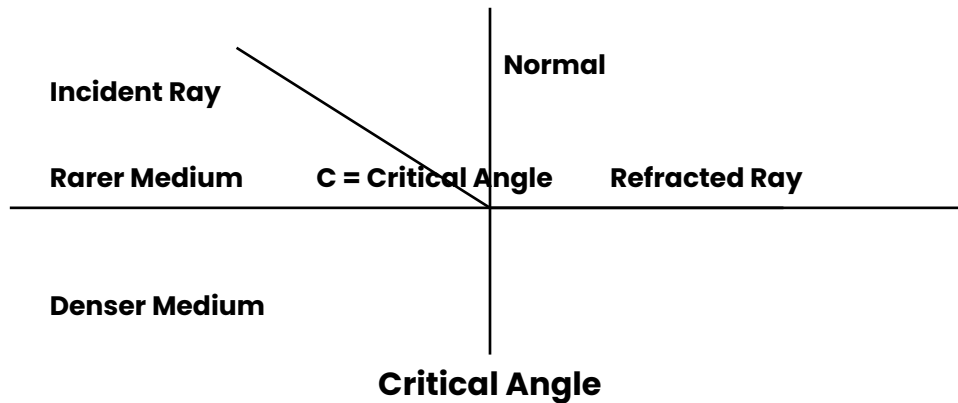


6. Critical Angle

Critical angle is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium becomes 90 degrees.

$$\sin C = 1 / n$$

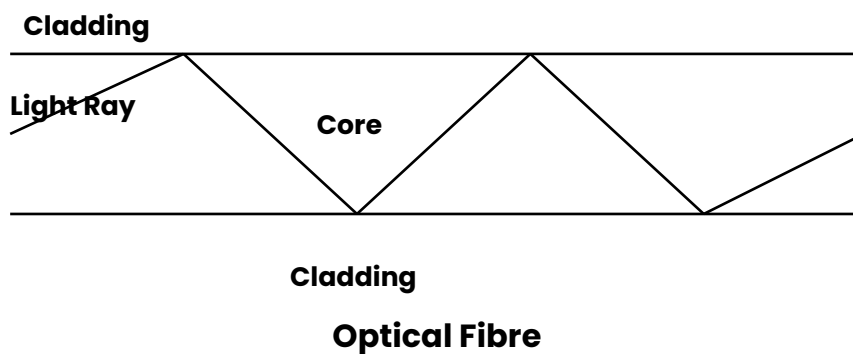
- C is the critical angle and n is the refractive index of the denser medium with respect to the rarer medium.



7. Optical Fibre

An optical fibre is a thin flexible transparent fibre that transmits light by repeated total internal reflection.

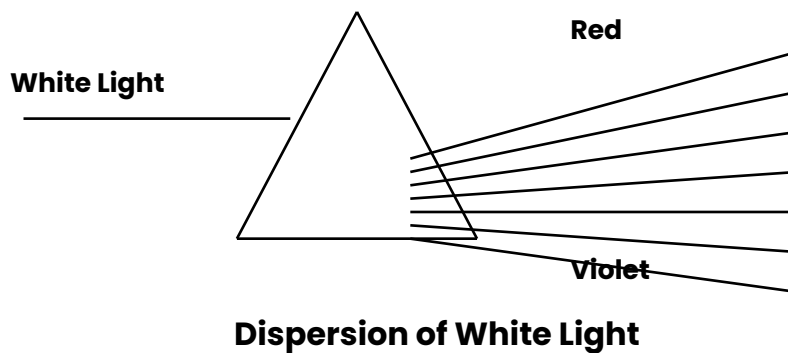
- The core has a higher refractive index than the cladding.
- Light travels through the core by total internal reflection.
- Optical fibres are used in communication systems and medical instruments.



8. Dispersion of Light

Dispersion is the splitting of white light into its constituent colours when it passes through a prism.

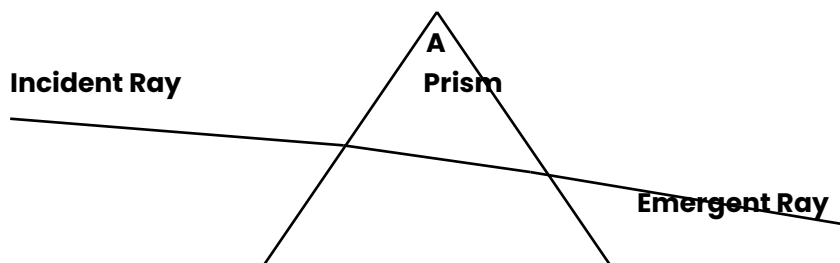
- White light consists of different colours having different wavelengths.
- Different colours travel with different speeds in a medium.
- Violet light deviates more than red light in a glass prism.



9. Prism

A prism is a transparent refracting medium bounded by two plane surfaces inclined at an angle.

- The angle between the two refracting surfaces is called the angle of prism.
- A light ray is deviated from its original direction after passing through a prism.



Refraction through Prism

10. Prism Formula

For a prism at minimum deviation, the refractive index is related to the angle of prism and minimum deviation.

$$n = \sin[(A + D_m) / 2] / \sin(A / 2)$$

- A is the angle of prism.
- D_m is the angle of minimum deviation.

11. Angular Dispersion & Dispersive Power

Angular dispersion is the angular separation between two colours of the spectrum produced by a prism.

$$\text{Angular Dispersion} = \delta_v - \delta_r$$

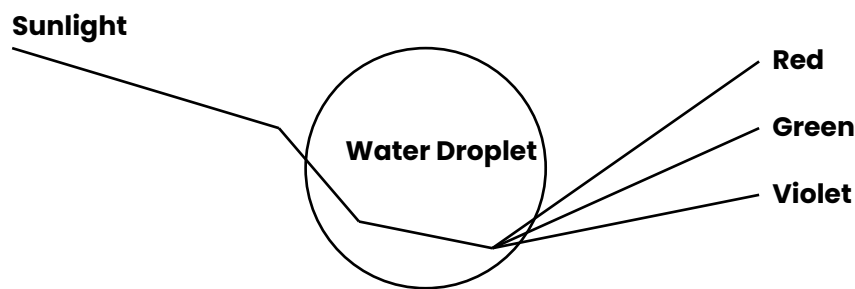
Dispersive power represents the ability of a material to disperse different colours of light.

$$\omega = (n_v - n_r) / (n_y - 1)$$

12. Rainbow

A rainbow is a natural spectrum formed by the dispersion, refraction and internal reflection of sunlight in water droplets.

- Sunlight enters a water droplet and is refracted and dispersed.
- The light undergoes internal reflection inside the droplet.
- The light is refracted again when it emerges from the droplet.



Rainbow Formation

13. Scattering of Light

Scattering is the phenomenon in which light is redirected in different directions by small particles present in a medium.

- Shorter wavelengths are scattered more strongly than longer wavelengths.
- Scattering of sunlight by atmospheric particles produces several natural optical effects.

14. Blue Colour of Sky

The sky appears blue because shorter wavelength blue light is scattered more strongly by the molecules of the atmosphere.

- Blue light is scattered much more than red light.
- The scattered blue light reaches our eyes from different directions.

15. Sunrise & Sunset

During sunrise and sunset, sunlight travels through a longer path in the atmosphere.

- Shorter wavelength colours are scattered away from the direct path.
- Red and orange light reach the observer more effectively.

- Therefore the Sun often appears reddish near the horizon.

16. Elementary Raman Effect

The Raman effect is the change in wavelength of a small fraction of light when it is scattered by molecules of a substance.

- The scattered light contains components with frequencies different from the incident light.
- The Raman effect provides information about molecular structure.

Important Formulae

$$n = c / v$$

$$n = \sin i / \sin r$$

$$\sin C = 1 / n$$

$$n = \sin[(A + D_m) / 2] / \sin(A / 2)$$

$$\text{Angular Dispersion} = \delta_v - \delta_r$$

$$\omega = (n_v - n_r) / (n_y - 1)$$

Quick Revision Points

- Refraction occurs because the speed of light changes from one medium to another.
- Frequency remains constant during refraction.

- Total internal reflection occurs only when light travels from a denser medium to a rarer medium and the angle of incidence exceeds the critical angle.
- Optical fibre works on the principle of total internal reflection.
- A prism can produce dispersion of white light.
- Violet light deviates more than red light in a prism.
- Blue colour of the sky is due to scattering of light.

MHT-CET Important Points / Numerical Focus

- Snell's law based numericals are important.
- Refractive index and speed of light based numericals are important.
- Critical angle and total internal reflection based questions are important.
- Prism formula and minimum deviation based numericals are important.
- Angular dispersion and dispersive power can be asked in numerical form.
- Optical fibre principle and applications are important.
- Questions based on rainbow, scattering, blue sky and sunrise or sunset are concept-oriented.